

FUNCTION

2. The domain of definition of $f(x) = \sec^{-1}(\cos^2 x)$ is
 (A) $m\pi, m \in \mathbb{I}$ (B) $\pi/2$
 (C) $\pi/4$ (D) none of these.
7. Which of the following function(s) from $f : A \rightarrow A$ are invertible, where $A = [-1, 1]$:
 (A) $f(x) = x/2$ (B) $g(x) = \sin(\pi x/2)$
 (C) $h(x) = |x|$ (D) $k(x) = x^2$
8. Solution of $0 < |x-3| \leq 5$ is
 (A) $[-2, 8]$ (B) $[-2, 3) \cup (3, 8]$ (C) $[-2, 3]$ (D) none of these
9. Solution of $\frac{(x-3)(x+5)(x-7)}{|x-4|(x+6)} \leq 0$ is
 (A) $(-6, -5] \cup [3, 7) \cup (4, 7)$ (B) $[3, 7]$
 (C) $(-6, -5]$ (D) $[3, 4) \cup (4, 7]$
13. If $f(x) = \sin^{-1}\left(\frac{x^2}{1+x^2}\right)$ then the range of $f(x)$ is
 (A) $[-\pi/2, \pi/2]$ (B) $[0, \pi/2]$
 (C) $[0, \pi/2)$ (D) $[-\pi/2, 0)$
14. If the period of $\frac{\sin(nx)}{\tan(x/n)}$, where $n \in \mathbb{I}$, is 6π , then
 (A) $n = 4$ (B) $n = -3$
 (C) $n = 3$ (D) none of these
19. Period of $|\sin 2x| + |\cos 8x|$ is:
 (A) $\pi/2$ (B) $\pi/8$
 (C) $\pi/16$ (D) None of these.
40. Range of $f(x) = \sin^{-1}\sqrt{x^2 + x + 1}$ is
 (A) $\left[\frac{\pi}{3}, \frac{\pi}{2}\right]$ (B) $\left[\frac{\pi}{3}, \frac{\pi}{4}\right]$
 (C) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$ (D) none of these
46. The function defined as $f : [0, \pi] \rightarrow [-1, 1]$, $f(x) = \cos x$ is
 (A) one-one onto (B) many-one onto
 (C) one-one into (D) many-one into
55. Period of the function $|\cos 2x|$ is
 (A) 2π (B) π
 (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$
57. If $f(x) = x^2$, $g(x) = \sqrt{x}$, then what is $g \circ f(x)$ is
 (A) $|x|$ (B) x
 (C) $-x$ (D) $-|x|$

63. If $f(x) = \frac{1}{1-x}$, then $f[f(f(x))]$ is
 (A) $x - 1$ (B) $1 - x$
 (C) x (D) $-x$
1. $\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$ is
 (A) 2 (B) -2 (C) 1/2 (D) -1/2
4. $f(x) = \begin{cases} ax^2 + bx + c, & |x| > 1 \\ x + 1, & |x| \leq 1 \end{cases}$. If $f(x)$ is continuous for all values of x , then;
 (A) $b = 1, a + c = 0$ (B) $b = 0, a + c = 2$
 (C) $b = 1, a + c = 1$ (D) none of these
5. The equation of the tangent to the curve $f(x) = 1 + e^{-2x}$ where it cuts the line $y = 2$ is
 (A) $x + 2y = 2$ (B) $2x + y = 2$
 (C) $x - 2y = 1$ (D) $x - 2y + 2 = 0$
10. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ is equal to
 (A) π (B) 1/4
 (C) 1/2 (D) 1
11. $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - 1}}{2x + 1}$ is equal to
 (A) 1 (B) 0
 (C) -1 (D) 1/2
12. $\lim_{x \rightarrow 0} \frac{x}{\tan^{-1} 2x}$ is equal to
 (A) 0 (B) 1/2
 (C) 1 (D) ∞
13. If $f(x) = (1 - x^n)^{1/n}$, $0 < x < 1$, n being an odd positive integer and $h(x) = f(f(x))$, then $h'\left(\frac{1}{2}\right)$ is equal to
 (A) 2^n (B) 2
 (C) $n \cdot 2^{n-1}$ (D) 1
17. The number of points of non differentiability for the function $f(x) = |\log |x||$ are
 (A) 2 (B) 4
 (C) 5 (D) 3
18. $\lim_{x \rightarrow 0} \frac{|x|}{x} =$
 (A) 0 (B) 1
 (C) -1 (D) doesn't exist
22. Function $f(x) = \tan x$ is continuous in the interval
 (A) $R - \left\{(2n+1)\frac{\pi}{2} : n \in I\right\}$ (B) $R - \{n\pi : n \in I\}$
 (C) R^+ (D) $R - \{0\}$

27. The value of $\lim_{x \rightarrow \infty} x \cos\left(\frac{\pi}{4x}\right) \sin\left(\frac{\pi}{4x}\right)$ is
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) 1 (D) π
31. The value of $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$ is
 (A) $\log_e \left(\frac{a}{b}\right)$ (B) $\log_e \left(\frac{b}{a}\right)$ (C) $\log_e (ab)$ (D) none of these
32. If $f(x) = \begin{cases} mx + 1, & x \leq \frac{\pi}{2} \\ \sin x + n, & x > \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$, then
 (A) $m = 1, n = 0$ (B) $m = \frac{n\pi}{2} + 1$ (C) $n = \frac{m\pi}{2}$ (D) $m = n = \frac{\pi}{2}$
34. The value of $\lim_{x \rightarrow \infty} \frac{\sqrt{1+x^4} - (1+x^2)}{x^2}$ is equal to
 (A) 0 (B) -1
 (C) 2 (D) 1
3. Area of the triangle formed by the positive x-axis and the normal and the tangent to $x^2 + y^2 = 4$ at $(1, \sqrt{3})$ is
 (A) $2\sqrt{3}$ sq. units (B) $\sqrt{3}$ sq. units
 (C) $4\sqrt{3}$ sq. units (D) none of these
4. A tangent to the curve $y = \frac{x^2}{2}$ which is parallel to the line $y = x$ cuts off an intercept from the y-axis is
 (A) 1 (B) -1/3
 (C) 1/2 (D) -1/2
5. A particle moves on a co-ordinate line so that its velocity at time t is $v(t) = t^2 - 2t$ m/sec. Then distance travelled by the particle during the time interval $0 \leq t \leq 4$ is
 (A) 4/3 (B) 3/4
 (C) 16/3 (D) 8/3
11. The greatest and least values of the function $f(x) = ax + b\sqrt{x} + c$, when $a > 0, b > 0, c > 0$ in the interval $[0,1]$ are
 (A) $a+b+c$ and c (B) $a/2, b\sqrt{2}+c, c$
 (C) $\frac{a+b+c}{\sqrt{2}}, c$ (D) None of these
12. The absolute minimum value of $x^4 - x^2 - 2x + 5$
 (A) is equal to 5 (B) is equal to 3
 (C) is equal to 7 (D) does not exist
13. Through the point $P(\alpha, \beta)$ where $\alpha\beta > 0$ the straight line $\frac{x}{a} + \frac{y}{b} = 1$ is drawn so as to form with co-ordinates axes a triangle of area S . If $ab > 0$, then the least value of S is
 (A) $2\alpha\beta$ (B) $1/2\alpha\beta$
 (C) $\alpha\beta$ (D) None of these